

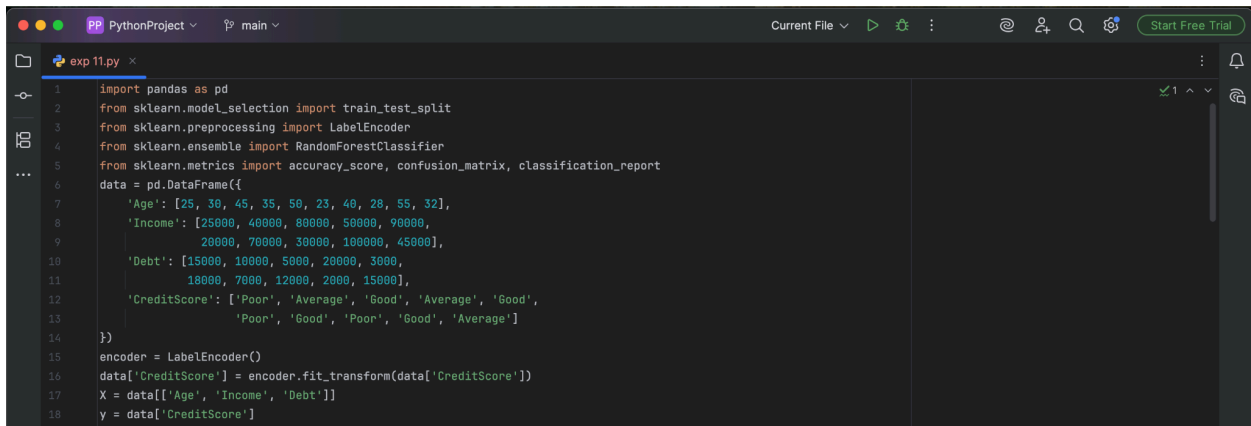
EXPERIMENT - 11

Aim: To develop a machine learning model to classify customers based on their credit score.

Procedure:

1. Import the required Python libraries.
2. Load the credit score dataset.
3. Clean the dataset and handle missing values.
4. Select relevant customer features.
5. Separate input features and credit score classes.
6. Split the data into training and testing sets.
7. Train a suitable classification model.
8. Predict the credit score class for test data.
9. Calculate the classification accuracy.
10. Display the predicted credit score categories and model performance.

Output:



```
1 import pandas as pd
2 from sklearn.model_selection import train_test_split
3 from sklearn.preprocessing import LabelEncoder
4 from sklearn.ensemble import RandomForestClassifier
5 from sklearn.metrics import accuracy_score, confusion_matrix, classification_report
6 data = pd.DataFrame({
7     'Age': [25, 30, 45, 35, 50, 23, 40, 28, 55, 32],
8     'Income': [25000, 40000, 80000, 50000, 90000,
9               20000, 70000, 30000, 100000, 45000],
10    'Debt': [15000, 10000, 5000, 20000, 3000,
11            18000, 7000, 12000, 2000, 15000],
12    'CreditScore': ['Poor', 'Average', 'Good', 'Average', 'Good',
13                  'Poor', 'Good', 'Poor', 'Good', 'Average']
14 })
15 encoder = LabelEncoder()
16 data['CreditScore'] = encoder.fit_transform(data['CreditScore'])
17 X = data[['Age', 'Income', 'Debt']]
18 y = data['CreditScore']
```

```
17 X = data[['Age', 'Income', 'Debt']]
18 y = data['CreditScore']
19 X_train, X_test, y_train, y_test = train_test_split(
20     *arrays: X, y, test_size=0.3, random_state=42
21 )
22 model = RandomForestClassifier(n_estimators=100, random_state=42)
23 model.fit(X_train, y_train)
24 y_pred = model.predict(X_test)
25 print("Predicted:", y_pred)
26 print("Actual:", y_test.values)
27 print("Accuracy:", accuracy_score(y_test, y_pred))
28 print("\nConfusion Matrix:")
29 print(confusion_matrix(y_test, y_pred))
30 print("\nClassification Report:")
31 print(classification_report(
32     y_test, y_pred,
33     target_names=encoder.classes_,
34     zero_division=0
35 ))
```

Run exp 11 x

PythonProject > exp 11.py 37:1 LF UTF-8 4 spaces Python 3.14 (PythonProject) 635 of 1500M

```
Run exp 11 x
...
/Users/moshika/PycharmProjects/PythonProject/.venv/bin/python /Users/moshika/PycharmProjects/PythonProject/exp 11.py
Predicted: [1 2 2]
Actual: [1 0 2]
Accuracy: 0.6666666666666666

Confusion Matrix:
[[0 0 1]
 [0 1 0]
 [0 0 1]]

Classification Report:

```

	precision	recall	f1-score	support
Average	0.00	0.00	0.00	1
Good	1.00	1.00	1.00	1
Poor	0.50	1.00	0.67	1
accuracy			0.67	3
macro avg	0.50	0.67	0.56	3
weighted avg	0.50	0.67	0.56	3

Result: The classification model successfully classified customers into their respective credit score categories and evaluated the prediction accuracy.